Flooding and Environmental Hazards Induced by Climate Change:

Flooding and Environmental Hazards Induced by Climate Change in a Rural Nile Delta Community:
A Case Study of Al-Dahria Village, Sherbin District, Dakahlia Governorate, Nile Delta, Egypt.

This research was prepared as part of a voluntary scientific contribution conducted within the activities of The Positive Event CIC.

Abstract

Climate change has intensified environmental hazards worldwide, with flooding representing one of the most severe threats in low-lying deltaic regions. The Nile Delta is among the most vulnerable deltas globally due to sea level rise, land subsidence, reduced sediment supply, and rising groundwater levels. This study presents an analytical assessment of flooding and associated environmental hazards induced by climate change, focusing on Al-Dahria Village, Sherbin District, Dakahlia Governorate, Egypt. Although Al-Dahria is not directly located on the Mediterranean coast, it is indirectly affected by delta-wide hydrological processes that extend inland. The study synthesizes peer-reviewed research based on remote sensing, GIS analysis, and vulnerability assessments conducted across the Nile Delta and adjacent coastal zones. By downscaling regional findings to the village scale, flood susceptibility, soil degradation, groundwater rise, and agricultural vulnerability are analyzed. The results indicate that climate change–driven processes affecting the Nile Delta coast propagate inland, increasing flood-related risks in rural settlements such as Al-Dahria. The study highlights the importance of integrated adaptation strategies that address both coastal and inland deltaic environments.

Keywords

Climate change; Flooding; Nile Delta; Sea level rise; Environmental hazards; Al-Dahria Village; Dakahlia Governorate

1. Introduction

Climate change is widely recognized as a major driver of environmental transformation, influencing hydrological systems, sea levels, and land stability. Among its most critical consequences is the increased frequency and severity of flooding, particularly in low-lying deltaic regions. These regions are highly sensitive due to minimal elevation above sea level, dense populations, and strong dependence on agriculture.

The Nile Delta represents one of the most climate-sensitive regions in the world. Large portions of the delta lie at elevations close to mean sea level, making them highly vulnerable to inundation under projected sea level rise scenarios (Hereher, 2010). In addition to eustatic sea level rise, the delta is experiencing significant land subsidence caused by sediment compaction, neotectonic activity, and reduced sediment delivery following the construction of the High Aswan Dam (Stanley & Clemente, 2017). These processes collectively intensify relative sea level rise and increase flood susceptibility.

Most existing studies on climate-induced flooding in Egypt focus on the northern coastal zone of the Nile Delta, including Alexandria, Kafr El-Sheikh, Damietta, Gamasa, and Ras El-Bar (Hassaan, 2013; Elshinnawy & Almaliki, 2021). However, inland rural settlements located away from the immediate coastline have received less attention, despite their exposure to indirect flooding mechanisms such as groundwater rise and drainage backwater effects.

Al-Dahria Village, located in Sherbin District within Dakahlia Governorate, represents a typical inland deltaic settlement. Although it is not directly exposed to coastal inundation, its location within the low-lying floodplain renders it vulnerable to climate change–driven hydrological processes. This study aims to analyze flooding and environmental hazards induced by climate change with a specific focus on Al-Dahria Village by synthesizing regional-scale studies and downscaling their findings to the local level.
This study also aims to support community-based environmental awareness initiatives by providing scientifically grounded insights relevant to vulnerable rural areas in the Nile Delta.

2. Study Area

Al-Dahria Village is situated in Sherbin District, Dakahlia Governorate, within the northeastern Nile Delta. Dakahlia Governorate occupies a central position in the deltaic floodplain and is characterized by flat topography, intensive agricultural land use, and dense rural settlements. Although the village is located inland, it remains hydrologically connected to the northern coastal zone of the Nile Delta.

The delta is generally flat, with elevations commonly ranging between −1 and 3 m relative to mean sea level (Hereher, 2010). Such low relief facilitates the inland propagation of climate-induced impacts, including rising groundwater levels and impaired drainage. Dakahlia Governorate lies within the eastern Nile Delta, where subsidence and groundwater rise have been documented as significant environmental processes (Stanley & Clemente, 2017).

Agricultural land dominates the surroundings of Al-Dahria Village, supported by an extensive irrigation and drainage network. The soils are primarily fine-grained alluvial deposits that are productive but sensitive to waterlogging, compaction, and salinization. Soil studies conducted in northern Dakahlia indicate widespread physical degradation associated with shallow groundwater and inefficient drainage (Abuzaid, 2018).

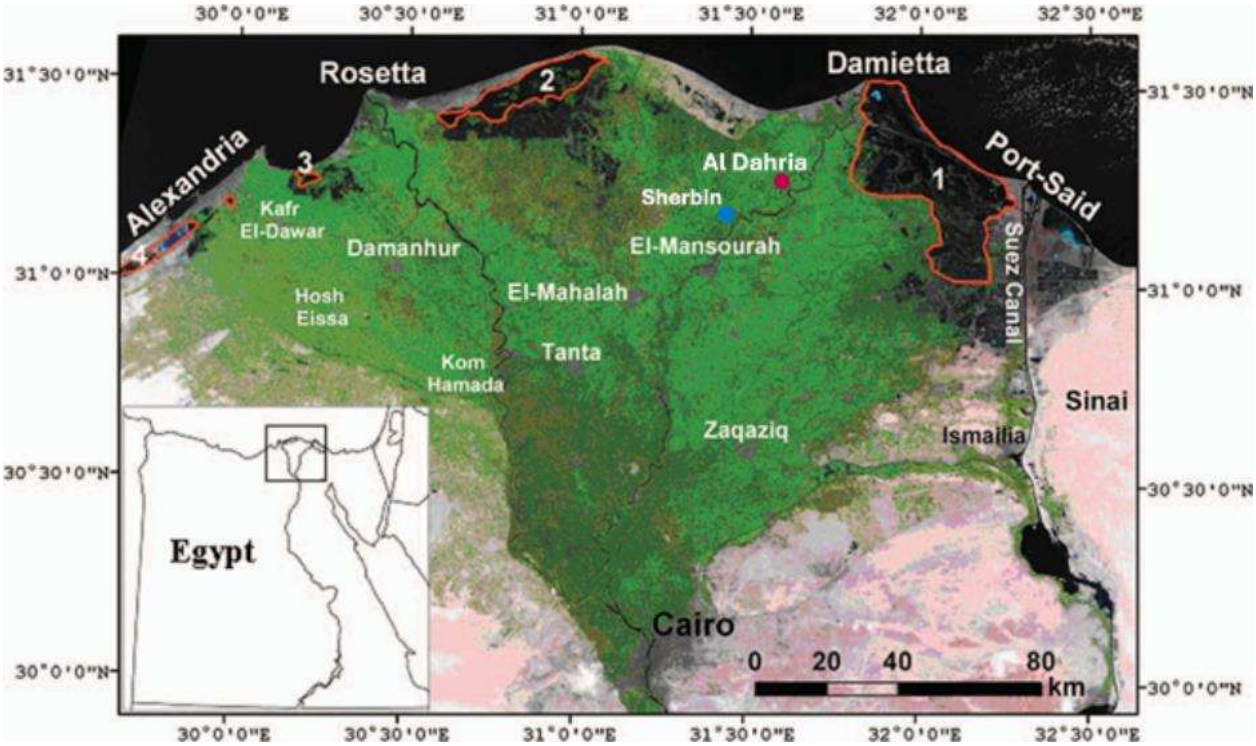

Figure 1. Location of Al-Dahriya Village within Sherbin District, Dakahlia Governorate, and its position relative to the Nile Delta and the Mediterranean coast.

3. Data and Methods

This study adopts an analytical approach based on the synthesis of peer-reviewed literature addressing flooding, sea level rise, land subsidence, groundwater dynamics, and soil degradation in the Nile Delta. Rather than generating new field data, the analysis relies on downscaling regional-scale findings to assess localized risks in Al-Dahria Village.

Primary sources include studies employing remote sensing, digital elevation models, and GIS-based vulnerability assessments (Hereher, 2010; Hassaan, 2013). These studies provide spatially explicit insights into areas vulnerable to inundation and environmental degradation under current and future climate scenarios.

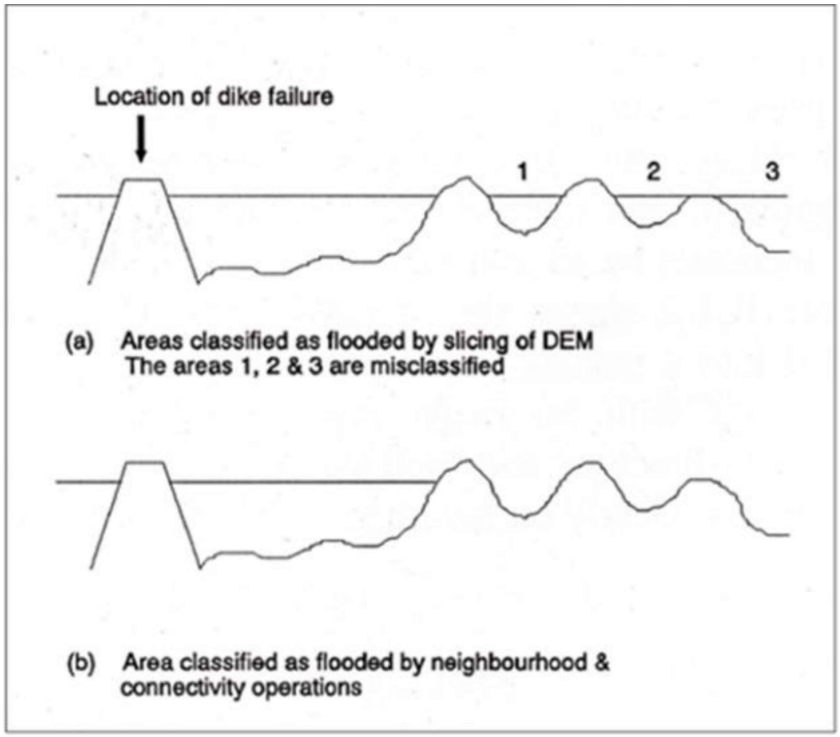

Figure 2: Conceptual illustration of flood mapping uncertainty using DEM-based slicing compared to neighbourhood and connectivity-based GIS approaches. *All figures were adapted from peer-reviewed sources.*

The methodology consists of three main steps. First, dominant climate change drivers of flooding in the Nile Delta are identified. Second, their relevance to inland deltaic environments is examined through studies documenting hydrological connectivity between coastal zones and interior floodplains (Elshinnawy & Almaliki, 2021). Third, these findings are contextualized for Al-Dahria Village based on its geographic and environmental setting.

4. Climate Change Drivers of Flooding in the Nile Delta

4.1 Sea Level Rise

Sea level rise is a primary driver of flooding in the Nile Delta. Large areas of the delta lie at low elevations, making them highly sensitive to even moderate increases in sea level (Hereher, 2010). Coastal vulnerability assessments demonstrate that sea level rise results in extensive flooding of agricultural and wetland areas (Hassaan, 2013).

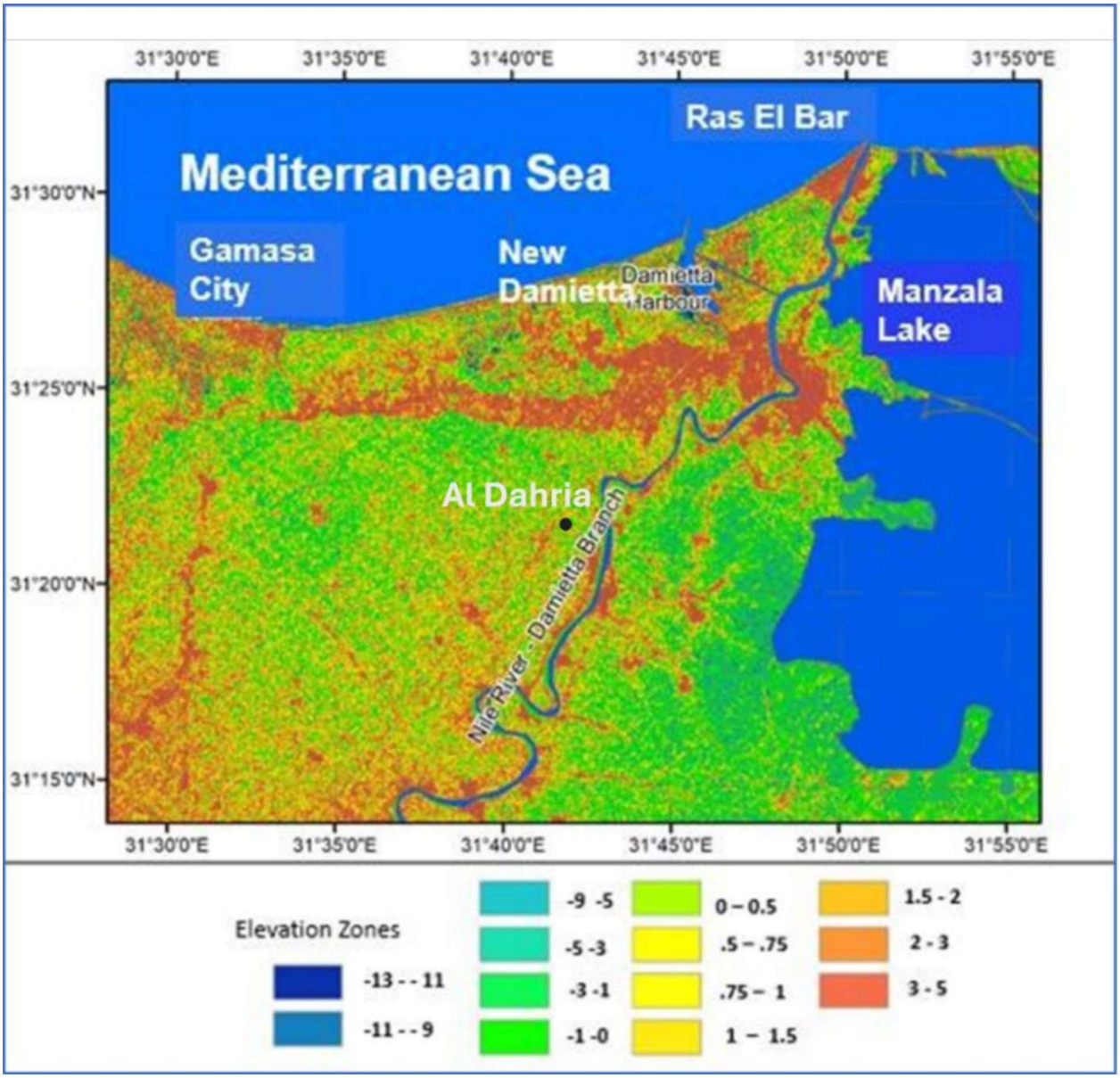

Figure 3. *Elevation zones of the northeastern Nile Delta derived from digital elevation model (DEM) analysis, highlighting the inland position of Al-Dahria Village relative to low-lying coastal areas and illustrating its susceptibility to indirect flooding processes*.

Although Al-Dahria Village is inland, sea level rise affects it indirectly by elevating downstream water levels and reducing drainage efficiency. Backwater effects propagate inland through canals and drains, increasing waterlogging risk (Elshinnawy & Almaliki, 2021).

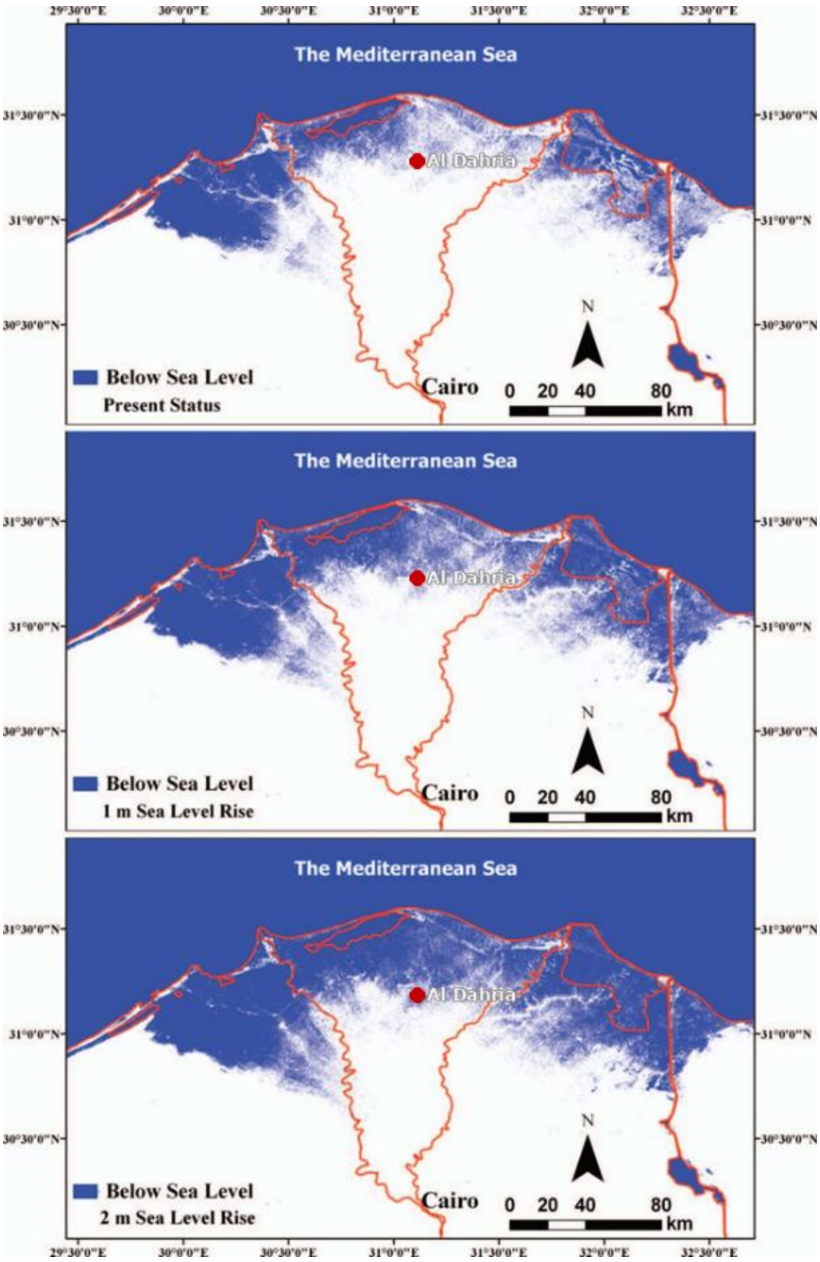

Figure 4. Elevation and sea level rise vulnerability of the Nile Delta based on remote sensing and GIS analyses.

4.2 Land Subsidence

Land subsidence significantly amplifies relative sea level rise in the Nile Delta. Subsidence rates in the eastern delta exceed global eustatic sea level rise in some areas, driven by sediment compaction and reduced sediment supply (Stanley & Clemente, 2017). Subsidence lowers land elevations, increasing flood susceptibility even in inland areas such as Dakahlia Governorate.

4.3 Reduced Sediment Supply

The construction of the High Aswan Dam has greatly reduced sediment delivery to the Nile Delta, limiting its capacity to naturally offset subsidence and sea level rise. Sediment starvation contributes to land degradation and reduced soil resilience to flooding (Stanley & Clemente, 2017).

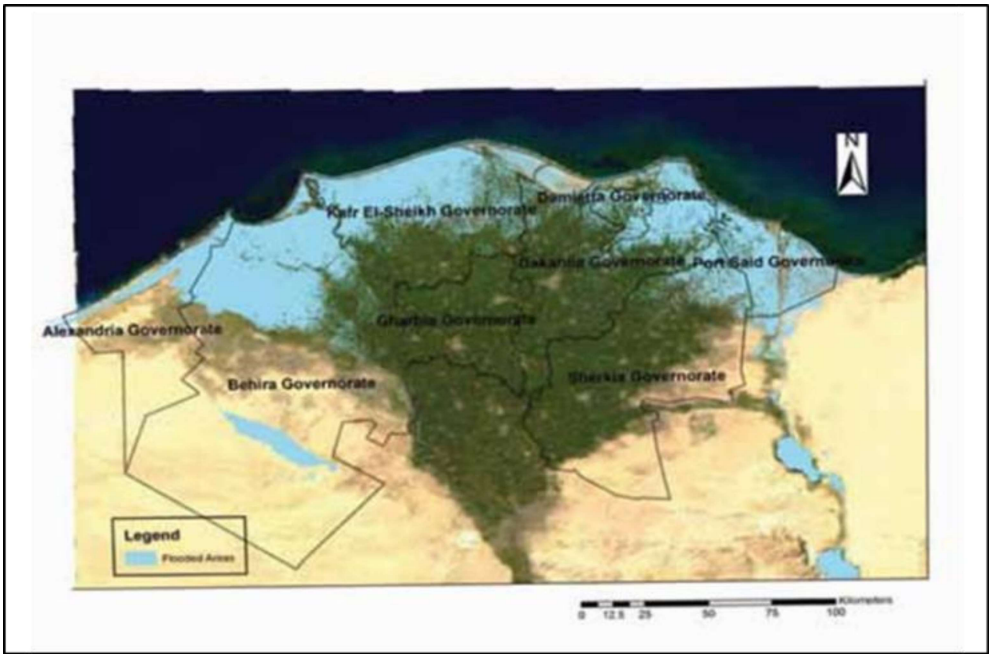

Figure 5. Potential overestimation of flood extent where non-hydrologically connected inland areas are classified as flooded, highlighting limitations of elevation-based mapping approaches. This limitation is particularly relevant when assessing inland villages such as Al-Dahria.

4.4 Rising Groundwater Levels

Rising groundwater levels represent a critical flooding mechanism in inland deltaic environments. Sea level rise and reduced drainage gradients lead to sustained high water tables, causing waterlogging and soil degradation (Elshinnawy & Almaliki, 2021). In Dakahlia Governorate, shallow groundwater has been identified as a major environmental hazard (Abuzaid, 2018).

5. Flooding and Environmental Hazards

Flooding in the Nile Delta includes both coastal inundation and inland waterlogging. While coastal flooding dominates near the Mediterranean shoreline, inland areas experience chronic waterlogging due to groundwater rise and poor drainage (Hassaan, 2013).

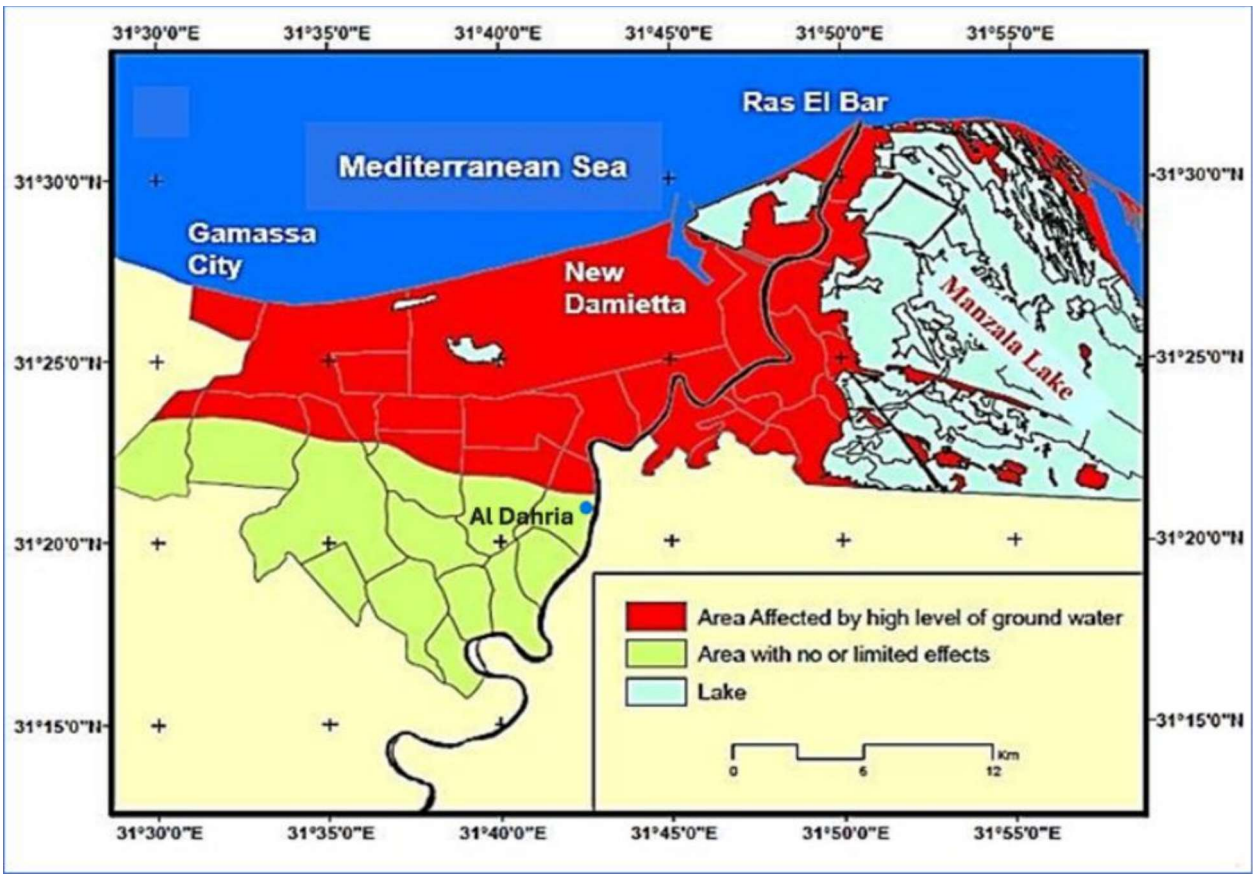

Figure 6. Flood risk and potential inundation zones in the northeastern Nile Delta, including Dakahlia and Damietta.

Soil degradation is a major consequence of prolonged flooding. Physical degradation processes such as compaction and structural collapse reduce infiltration and exacerbate surface water accumulation. Soil studies in northern Dakahlia indicate moderate to severe degradation driven by waterlogging (Abuzaid, 2018).

6. Implications for Al-Dahria Village

Al-Dahria Village faces increasing flood-related risks primarily through slow-onset processes. Rising groundwater levels and subsidence increase the likelihood of persistent waterlogging, negatively affecting agriculture and soil quality. Although catastrophic flooding is unlikely, chronic environmental stress threatens long-term sustainability.

7. Discussion

The case of Al-Dahria Village demonstrates that climate change impacts extend beyond coastal zones into inland deltaic environments. Similar patterns of groundwater rise and soil degradation have been documented in other eastern Nile Delta regions (Hassaan, 2013; Abuzaid, 2018). While uncertainties remain regarding future climate scenarios, the convergence of evidence indicates increasing vulnerability of inland villages.

8. Adaptation and Risk Reduction Measures

Adaptation strategies should focus on improving drainage efficiency, regulating irrigation practices, and enhancing soil management. Integrated water management approaches can reduce groundwater rise and mitigate waterlogging (Elshinnawy & Almaliki, 2021). Policy frameworks should address inland settlements alongside coastal protection.

9. Conclusion

This study provides an analytical assessment of climate change–induced flooding and environmental hazards in Al-Dahria Village within the Nile Delta. The findings indicate that inland deltaic settlements are increasingly affected by indirect flooding mechanisms driven by sea level rise, land subsidence, and groundwater dynamics. Addressing these risks requires integrated adaptation strategies that consider both coastal and inland dimensions of climate vulnerability. These findings can support local awareness initiatives and community-level adaptation planning.

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